

Model	Setting	8k		16k		32k		64k		Avg.	
		SFT	+RL	SFT	+RL	SFT	+RL	SFT	+RL	SFT	+RL
Untrained base models											
8B	DTC	1.3		0.0		0.0		0.0		0.3	
	Hybrid	1.3		0.0		0.0		0.0		0.3	
	PTC	0.0		0.0		0.0		0.0		0.0	
14B	DTC	1.3		0.0		0.0		0.0		0.3	
	Hybrid	2.7		0.0		0.0		0.0		0.7	
	PTC	0.0		0.0		0.0		0.0		0.0	
Trained baselines under the PTC harness											
AWM-8B	PTC harness	0.0		0.0		0.0		0.0		0.0	
AWM-14B	PTC harness	0.0		0.0		0.0		0.0		0.0	
EnvFactory-8B	PTC harness	0.0		0.0		0.0		0.0		0.0	
EnvScaler-8B	PTC harness	1.3		0.0		0.0		0.0		0.3	
Training paradigms, scaled environments (v_0 – v_3)											
8B	DTC	5.3	5.3	6.7	6.7	0.0	1.3	0.0	2.7	3.0	4.0
	DTC, memory tool	6.7	4.0	1.3	4.0	0.0	1.3	0.0	1.3	2.0	2.7
	DTC, PTC harness	16.0	20.0	10.7	10.7	8.0	8.0	1.3	4.0	9.0	10.7
	Hybrid, DTC harness	10.7	13.3	6.7	8.0	6.7	4.0	4.0	4.0	7.0	7.3
	Hybrid	22.7	28.0	24.0	28.0	13.3	16.0	10.7	14.7	17.7	21.7
	PTC, DTC harness	5.3	17.3	4.0	5.3	0.0	0.0	0.0	1.3	2.3	6.0
	PTC, memory tool	10.7	16.0	4.0	13.3	0.0	6.7	0.0	1.3	3.7	9.3
	PTC	24.0	34.7	20.0	24.0	20.0	21.3	20.0	17.3	21.0	24.3
14B	DTC	8.0	14.7	12.0	13.3	4.0	4.0	0.0	4.0	6.0	9.0
	DTC, memory tool	16.0	20.0	10.7	13.3	2.7	4.0	4.0	4.0	8.3	10.3
	DTC, PTC harness	25.3	28.0	24.0	24.0	10.7	17.3	13.3	13.3	18.3	20.7
	Hybrid, DTC harness	10.7	26.7	10.7	12.0	6.7	13.3	4.0	2.7	8.0	13.7
	Hybrid	26.7	38.7	25.3	30.7	14.7	33.3	12.0	22.7	19.7	31.3
	PTC, DTC harness	12.0	23.7	5.3	18.7	4.0	8.0	2.7	5.3	6.0	13.9
	PTC, memory tool	21.3	24.0	13.3	14.7	6.7	9.3	1.3	5.3	10.7	13.3
	PTC	32.0	41.3	30.7	40.0	33.3	41.3	28.0	30.7	31.0	38.3
Initial environments (v_0) only											
8B	DTC	5.3	8.0	2.7	2.7	1.3	2.7	1.3	0.0	2.7	3.3
	DTC, PTC harness	5.3	4.0	4.0	4.0	2.7	4.0	1.3	2.7	3.3	3.7
	PTC, DTC harness	8.0	13.3	2.7	8.0	1.3	4.0	1.3	2.7	3.3	7.0
	PTC	13.3	21.3	13.3	18.7	14.7	14.7	4.0	9.3	11.3	16.0
14B	DTC	8.0	13.3	10.7	8.0	4.0	4.0	0.0	1.3	5.7	6.7
	DTC, PTC harness	16.0	20.0	17.3	20.0	6.7	14.7	1.3	10.7	10.3	16.3
	PTC, DTC harness	18.7	20.0	10.7	13.3	5.3	6.7	2.7	2.7	9.3	10.7
	PTC	24.0	25.3	26.7	26.7	21.3	26.7	13.3	16.0	21.3	23.7
Server-level error detection disabled											
8B	PTC	29.3		18.7		17.3		12.0		19.3	
14B	PTC	42.0		40.0		29.3		26.7		34.5	